

Q9 → case study: vehicle management

CLASS Vehicle

DATA

VehicleNO

OwnerName

METHOD Vehicle(VehicleNO, OwnerName)

this.VehicleNO ← VehicleNO

this.OwnerName ← OwnerName

END METHOD

METHOD display()

PRINT "vehicle No: " + VehicleNO

PRINT "Owner Name: " + OwnerName

END METHOD

END CLASS

CLASS Car INHERITS Vehicle

DATA

FuelType

Method Car(VehicleNO, OwnerName, FuelType)

SUPER(VehicleNO, OwnerName)

this.FuelType ← FuelType

END METHOD

END CLASS

CLASS Truck INHERITS Vehicle

DATA

loadCapacity

METHOD Truck(VehicleNO, OwnerName,

loadCapacity)

SUPER(VehicleNO, OwnerName)

this.loadCapacity ← loadCapacity

END METHOD

METHOD display()

SUPER.display()

PRINT "Load Capacity: " + loadCapacity.

END METHOD

END CLASS

BEGIN

READ N

Vehicle Vehicles[N]

FOR i ← 0 TO N-1

READ type

READ vehicleNo, ownerName

IF type == "Car"

READ fuelType

Vehicles[i] ← New Car(vehicleNo,
ownerName, fuelType)

ELSE IF type == "Truck"

READ loadCapacity

Vehicles[i] ← New Truck(vehicleNo,
ownerName, loadCapacity)

ENDIF

END FOR

FOR i ← 0 TO N-1

Vehicles[i].display()

ENDFOR

END